科研课堂课程作业

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题目1. 验证方程(1)通过变换(2)变换成方程(3):

$$\frac{\partial \eta}{\partial t} = \frac{3}{2}\sqrt{\frac{g}{l}}\frac{\partial}{\partial x}(\frac{1}{2}\eta^2 + \frac{2}{3}\alpha\eta + \frac{1}{3}\sigma\frac{\partial^2\eta}{\partial x^2}) \tag{1}$$

$$t' = \frac{1}{2}\sqrt{\frac{g}{l\sigma}}t, x' = -\frac{x}{\sqrt{\sigma}}, u = \frac{1}{2}\eta + \frac{1}{3}\alpha \tag{2}$$

$$\frac{\partial u}{\partial t} + 6u\frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0. \tag{3}$$

解答. 我们有

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial t'}\frac{\mathrm{d}t'}{\mathrm{d}t} = \frac{\partial}{\partial t'}\cdot\frac{1}{2}\sqrt{\frac{g}{l\sigma}}$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial x'}\frac{\mathrm{d}x'}{\mathrm{d}x} = \frac{\partial}{\partial x'}\cdot(-\frac{1}{\sqrt{\sigma}})$$

$$\frac{\partial u}{\partial t} = \frac{1}{2}\frac{\partial \eta}{\partial t}, \frac{\partial^2 u}{\partial x^2} = \frac{1}{2}\frac{\partial^2\eta}{\partial x^2}$$

代入之后有

$$\begin{aligned}\frac{\partial \eta}{\partial t} &= 2\frac{\partial u}{\partial t}\\ &= \sqrt{\frac{g}{l\sigma}}\frac{\partial u}{\partial t'}\end{aligned}$$

$$\begin{aligned}&\frac{1}{2}\eta^2 + \frac{2}{3}\alpha\eta + \frac{1}{3}\sigma\frac{\partial^2\eta}{\partial x^2}\\ =&\frac{1}{2}(2u - \frac{2}{3}\alpha)^2 + \frac{2}{3}\alpha(2u - \frac{2}{3}\alpha) + \frac{2}{3}\frac{\partial^2 u}{\partial x'^2}\\ =&2u^2 - \frac{2}{9}\alpha^2 + \frac{2}{3}\frac{\partial^2 u}{\partial x'^2}\end{aligned}$$

$$\begin{aligned}&\frac{3}{2}\sqrt{\frac{g}{l}}\frac{\partial}{\partial x}(2u^2-\frac{2}{9}\alpha^2+\frac{2}{3}\frac{\partial^2 u}{\partial x'^2})\\=&-\frac{3}{2}\sqrt{\frac{g}{l\sigma}}(\frac{\partial(2u^2)}{\partial x'}+\frac{2}{3}\frac{\partial^3 u}{\partial x'^3})\\=&-6\sqrt{\frac{g}{l\sigma}}u\frac{\partial u}{\partial x'}-\sqrt{\frac{g}{l\sigma}}\frac{\partial^3 u}{\partial x'^3}\end{aligned}$$

去掉“′”标记之后,即可得到标准KdV方程:

$$\frac{\partial u}{\partial t}+6u\frac{\partial u}{\partial x}+\frac{\partial^3 u}{\partial x^3}=0.$$

题目2. 验证KdV方程有行波解:

$$u(x,t)=\frac{c}{2}\mathrm{sech}^2(\frac{\sqrt{c}}{2}(x-ct))$$

解答. 计算可得

$$\frac{\partial^3 u}{\partial x^3}=\frac{1}{2}c^{\frac{5}{2}}\tanh(\frac{\sqrt{c}}{2}(x-ct))\mathrm{sech}^2(\frac{\sqrt{c}}{2}(x-ct))(2\mathrm{sech}^2(\frac{\sqrt{c}}{2}(x-ct))-\tanh^2(\frac{\sqrt{c}}{2}(x-ct)))$$

$$6u\frac{\partial u}{\partial x}=-\frac{3}{2}c^{\frac{5}{2}}\tanh(\frac{\sqrt{c}}{2}(x-ct))\mathrm{sech}^4(\frac{\sqrt{c}}{2}(x-ct))$$

$$\frac{\partial u}{\partial t}=\frac{1}{2}c^{\frac{5}{2}}\tanh(\frac{\sqrt{c}}{2}(x-ct))\mathrm{sech}^2(\frac{\sqrt{c}}{2}(x-ct))$$

同时有

$$\tanh^2(x)=1-\mathrm{sech}^2(x)$$

代入即可验证行波解是KdV方程的解.

题目3. 画出行波解的图形

解答. 以下给出的为$c=2$ 时的图形

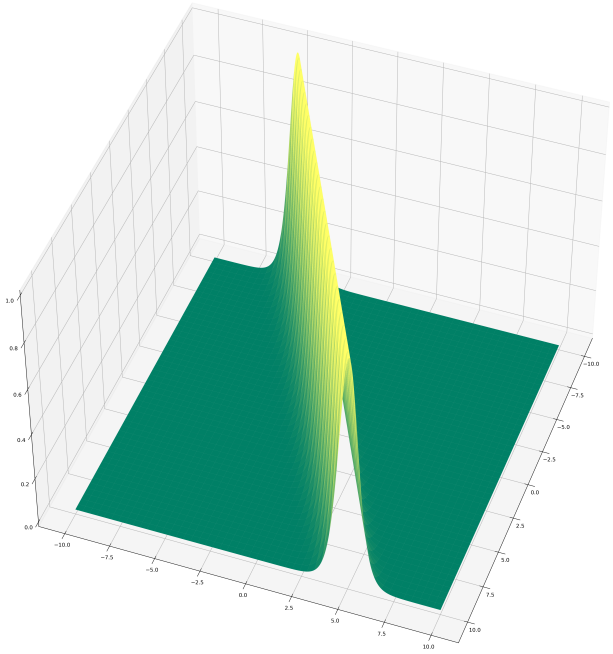


图 1: $c = 2$ 时的行波解